

15 Statutory Rules & Deterministic Scoring Engine Architecture

Comprehensive Guide to Label Verification, Rule Classification, Mathematical Scoring (0–100), and Defensibility

ENGINE: v1.0 Production

1. Executive Architecture: The 2-Stage Pipeline

The foundational engineering decision of this compliance platform is the **strict separation** between Multimodal Vision AI (OCR) and Statutory Legal Judgment. Large Language Models should never directly declare whether a product is legally compliant, because neural networks exhibit stochastic output drift, hallucinate absent text, and cannot be defended in official audits. Our platform divides the task into two distinct, auditable stages:

STAGE 1: Vision OCR & Transcription (Google Gemini Multimodal API, Temperature = 0)	STAGE 2: Deterministic Rule Engine (Pure Python 3, Zero AI, Zero Randomness)
• Sole Objective: Answer "What text is printed on this package?" • Strictly limited to transcription of 12 declaration fields. • Governed by negative prompt constraints (never infer, guess, or invent). • Output validated against strict Pydantic JSON schema. • Forbidden from making legal compliance determinations.	• Sole Objective: Answer "Does that text satisfy statutory rules?" • Evaluates 15 codified rules from the 2011 Gazette notification. • Executes deterministic regex pattern matching & presence logic. • Computes weighted compliance score out of 100. • 100% reproducible: the same text always produces the exact same verdict.

2. Mathematical Scoring Engine (How Products are Scored Out of 100)

The compliance score is not an arbitrary estimate. It is an exact **statutory ratio** calculated from the weight of legal declarations that passed against the total weight of declarations that could be evaluated:

THE COMPLIANCE SCORING FORMULA

$$\text{Compliance Score} = \text{round} \left[100 \times \left(\frac{\sum \text{Passed Rule Weights}}{\sum \text{Scoreable Rule Weights}} \right) \right]$$

Where *Scoreable Rules* = All rules yielding **PASS** or **FAIL**.

Rules yielding **REVIEW** (manual physical checks) or **NOT_APPLICABLE** are strictly excluded from both numerator and denominator.

Key Principles Governing the Scoring Formula:

- **Weight Distribution:** Mandatory presence rules carry a weight of **2 points** each (omitting a mandatory declaration violates the core act), while format and conditional rules carry a weight of **1 point** each.
- **Unbiased Exclusions:** Physical verification items (e.g. Rule 9 letter height) cannot be measured from a 2D smartphone photo. Instead of fabricating numbers, they receive **REVIEW** (weight 0) and are excluded from the score. A package is never penalized for what an image cannot answer.
- **Double-Penalty Prevention:** If a declaration is missing entirely (e.g. no MRP on label), it receives **FAIL** under the presence check (LM006, losing 2 points). However, its corresponding format check (LM008) automatically evaluates to **NOT_APPLICABLE** rather than a second **FAIL**. This prevents one missing declaration from deducting double points.

3. The 4 Check Categories (Mechanisms of Verification)

Every statutory rule belongs to one of four check types handled by dedicated verification subroutines:

Check Type	Rules Count	Weight	Evaluation Mechanism & Statutory Handling
presence	7 Rules (LM001–LM007)	2 each	Inspects whether mandatory statutory text is present. Supports legal alternative keys via <i>alt_fields</i> (e.g., LM001 accepts manufacturer name, packer name, OR importer name). Yields PASS if text is found; yields FAIL if null.
format	4 Rules (LM008–LM011)	1 each	Executes statutory regular expressions against detected text. If the field is missing, returns NOT_APPLICABLE (prevents double penalty). If present and compliant with prescribed gazette syntax, returns PASS ; if non-compliant (e.g. "100 gms"), returns FAIL with escalation.

conditional_ presence	1 Rule (LM012)	1	Evaluates rules that apply only under specific trigger conditions. For imported commodities (LM012), the check is NOT APPLICABLE unless an importer name is detected. Once an importer is named, Country of Origin becomes strictly mandatory.
manual	3 Rules (LM013–LM015)	0 (Unscored)	Covers physical package properties that require millimeter calipers or schedule classification (Rule 9 letter height, display prominence, pack sizes). Always returns REVIEW . Never guesses numbers, maintaining legal integrity.

4. Master Catalog of All 15 Implemented Rules

The table below presents all 15 rules codified in the compliance engine according to the Legal Metrology (Packaged Commodities) Rules, 2011:

ID	Gazette Ref	Statutory Requirement & Legal Scope	Check Type	Wt.	Escalates?
LM001	Rule 6(1)(a)	Manufacturer / Packer / Importer Name: Must declare name of manufacturer, packer, or importer.	presence	2	No
LM002	Rule 6(1)(a)	Complete Address: Declaration must include complete physical address, not only a company name.	presence	2	No
LM003	Rule 6(1)(b)	Generic Name of Commodity: Must state common or generic name (what product is, not just brand).	presence	2	No
LM004	Rule 6(1)(c)	Net Quantity Declaration: Must state net quantity in terms of standard metric weight/measure/count.	presence	2	No
LM005	Rule 6(1)(d)	Date of Manufacture / Packing: Must declare month and year of manufacture, pre-packing, or import.	presence	2	No
LM006	Rule 6(1)(e)	Retail Sale Price (MRP): Must declare the maximum retail sale price of the package.	presence	2	No
LM007	Rule 6(1)	Consumer Care Details: Must state contact details for consumer complaints.	presence	2	No
LM008	Rule 6(1)(e)	MRP Inclusive of All Taxes: Price must include prescribed wording <i>"inclusive of all taxes"</i> .	format	1	Yes
LM009	Rule 8	Standard Metric Unit Symbol: Net quantity must use prescribed symbols (g, kg, ml, l). Rejects <i>"gms"</i> , <i>"gm"</i> , <i>"ltr"</i> .	format	1	Yes
LM010	Rule 6(1)(d)	Readable Month & Year: Date must identify valid month & year (e.g. 06/2026, JUN 2026). Rejects bare <i>"2026"</i> .	format	1	Yes
LM011	Rule 6(1)	Consumer Care Contact Route: Must include actionable telephone number or email address, not only a department.	format	1	Yes
LM012	Rule 6	Country of Origin on Imports: Mandated when an importer is declared. If domestic, evaluated as Not Applicable.	conditional	1	Yes
LM013	Rule 9 & Sch. II	Letter & Numeral Height: Prescribes minimum physical millimeter font height based on PDP area. Requires physical check.	manual	0	No
LM014	Rule 7	Principal Display Panel Prominence: Declarations must be conspicuous and grouped on the display panel.	manual	0	No
LM015	Second Schedule	Permitted Standard Pack Sizes: Certain commodities must be sold only in prescribed schedule quantities.	manual	0	No

5. The 3 Legal Verdicts & The Escalation Tree

The engine classifies every inspection into one of three statutory verdicts based on a defensible decision hierarchy:

Verdict	Statutory Meaning & Trigger Conditions	Legal Rationale & Defense
COMPLIANT	- All scoreable presence and format rules evaluated to PASS. - Image quality is verified as good and all text is legible. - Score = 100 / 100.	The product satisfies all automatically verifiable requirements under the 2011 Rules. Manual physical verification items (LM013–LM015) are noted for physical audit.
NEEDS_REVIEW	1 or 2 mandatory declarations missing (absence of evidence). - OR photo is marked blurry or partial. - Typical Score: 88 / 100.	Absence of evidence is not evidence of a defect. A smartphone photo only captures 1 or 2 faces. Missing declarations might be printed on unphotographed panels. The system flags this for officer review rather than falsely declaring a legal violation.
POTENTIAL_VIOLATION	A visible defect is detected (e.g. "100 gms", missing tax wording on MRP, bad date). - OR ≥ 3 mandatory declarations are missing simultaneously. - Typical Score: ≤ 71 / 100.	When an illegal format is visibly printed on the package, photographic proof of a statutory violation exists. Similarly, when 3+ mandatory items are absent, an incomplete label is statistically far more probable than an incomplete photo.

6. Worked Demonstration Cases & Mathematical Walkthroughs

The following four real-world scenarios illustrate exactly how weights are summed and scores are derived across diverse label conditions:

Scenario / Sample Product	Engine Findings & Rule Results	Mathematical Calculation	Final Verdict
Example 1: Fully Compliant Product (Krisp Gold Biscuit, 200g)	- All 7 presence rules: PASS (14 pts) - All 4 format rules: PASS (4 pts) - Conditional origin: NOT_APPLICABLE (Domestic) 3 Manual rules: REVIEW (Excluded)	Passed Weight = 14 + 4 = 18 Scoreable Weight = 18 Score = 100 × (18 / 18) = 100	COMPLIANT (Score: 100)
Example 2: Missing MRP Declaration (Daily Choice Basmati, 1kg)	6 presence rules: PASS (12 pts) - LM006 (MRP presence): FAIL (0 / 2 pts) - LM008 (MRP format): NOT_APPLICABLE (Double penalty prevented) 3 format rules: PASS (3 pts)	Passed Weight = 12 + 3 = 15 Scoreable Weight = 15 + 2 = 17 Score = 100 × (15 / 17) = 88	NEEDS REVIEW (Score: 88)
Example 3: Non-Standard Metric Unit (AgroPure Sooji, declared as "500 gms")	- All 7 presence rules: PASS (14 pts) 3 format rules: PASS (3 pts) - LM009 (Unit Symbol): FAIL (0 / 1 pt, "gms" is illegal symbol) - Visible defect escalates to violation	Passed Weight = 14 + 3 = 17 Scoreable Weight = 17 + 1 = 18 Score = 100 × (17 / 18) = 94	POTENTIAL VIOLATION (Score: 94)
Example 4: Multiple Statutory Defects (QuickBite Noodles)	5 presence rules: PASS (10 pts) - LM002 (Address) & LM007 (Care): FAIL (0 / 4 pts) - LM009 (Unit): FAIL (0 / 1 pt, "75 gm") 2 format rules: PASS (2 pts)	Passed Weight = 10 + 2 = 12 Scoreable Weight = 12 + 4 + 1 = 17 Score = 100 × (12 / 17) = 71	POTENTIAL VIOLATION (Score: 71)

7. Legal & Regulatory Defensibility Summary

When presenting this system to government officials, statutory inspectors, or hackathon judges, the following four core arguments prove the system's mathematical and legal soundness:

- **Zero Non-Deterministic Drift:** By eliminating AI from the scoring layer, the exact same label will generate the exact same score (e.g. 88 or 100) regardless of whether it is evaluated today, next month, or in a courtroom demonstration.
- **Direct Gazette Citation:** Every finding explicitly references its parent section in the 2011 Gazette (Rule 6(1)(a) through Rule 9 and Schedules), providing immediate statutory authority for every inspection certificate.
- **Safe Fallback for Incomplete Angles:** The engine acknowledges physical photography limits and distinguishes between "not seen on this photograph" (Needs Review) and "visibly printed defect" (Potential Violation).
- **Extensive Unit Test Validation:** The implementation is backed by a 58-assertion deterministic test suite in backend/test_compliance.py that validates all 15 rules, regex pattern edge cases, and scoring calculations with a 100% pass rate.